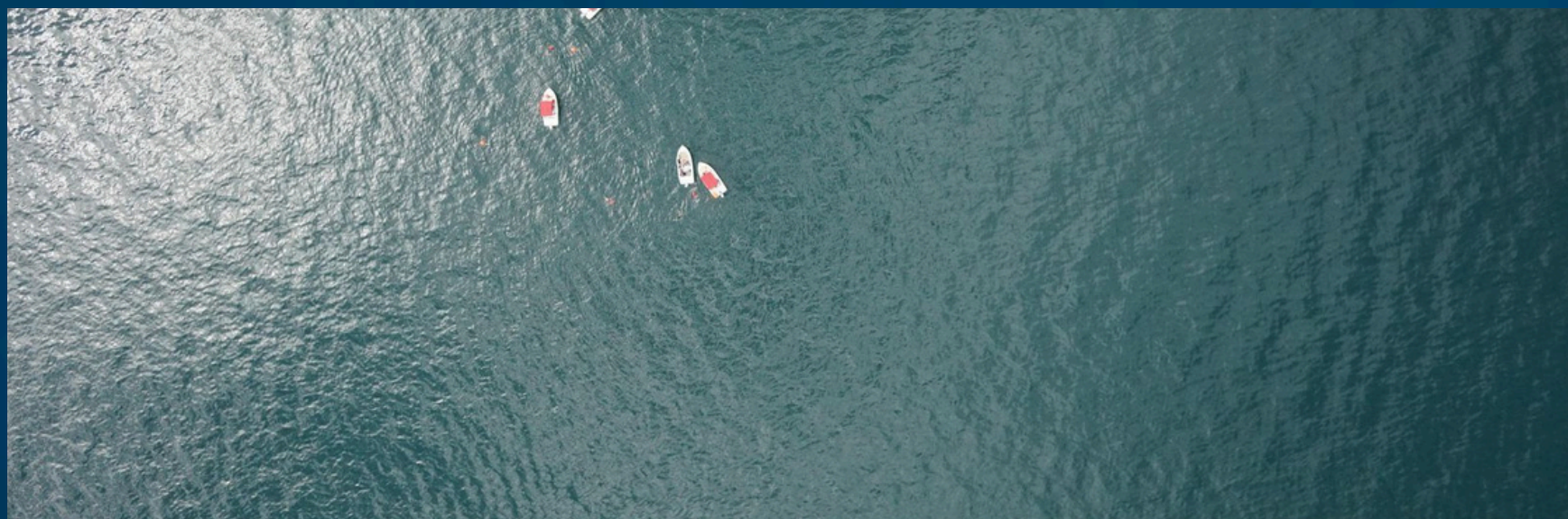


STORMVISION

PRESENTER: TOMER ATIA, SHALEV COHEN, NOAM HADAD

**MARITIME OBJECT DETECTION FROM AERIAL DRONE IMAGERY IN
STORMY CONDITIONS USING SYNTHETIC DATA AUGMENTATION**

ORIGINAL IMAGE



SYNTHETIC IMAGE



Review & Refined Project Definition

Maritime search-and-rescue operations rely heavily on drone-based detection of swimmers, boats, and other small objects.

The Data Gap:

There is currently NO publicly available labeled dataset of aerial drone footage capturing maritime objects in stormy sea conditions.

INPUT

Calm-sea drone images with
COCO annotations
SeaDronesSee Dataset:
8,262 train | 1,547 test



OUTPUT

Detection: Bounding boxes for 5 object classes
swimmer | boat | jetski | life_saving_appliances | buoy
Trained Model: YOLOv8s robust to stormy
conditions

Project Achievements and Novelty

PROJECT ACHIEVEMENTS

- End-to-end pipeline for maritime object detection under adverse weather conditions
- Synthetic stormy background generation pipeline:
Image → masks+depth map → SDXL Inpaint+ControlNet → Composite
- Controlled comparison: COCO Pretrained vs Real-Only vs Real+Synthetic (Mixed)

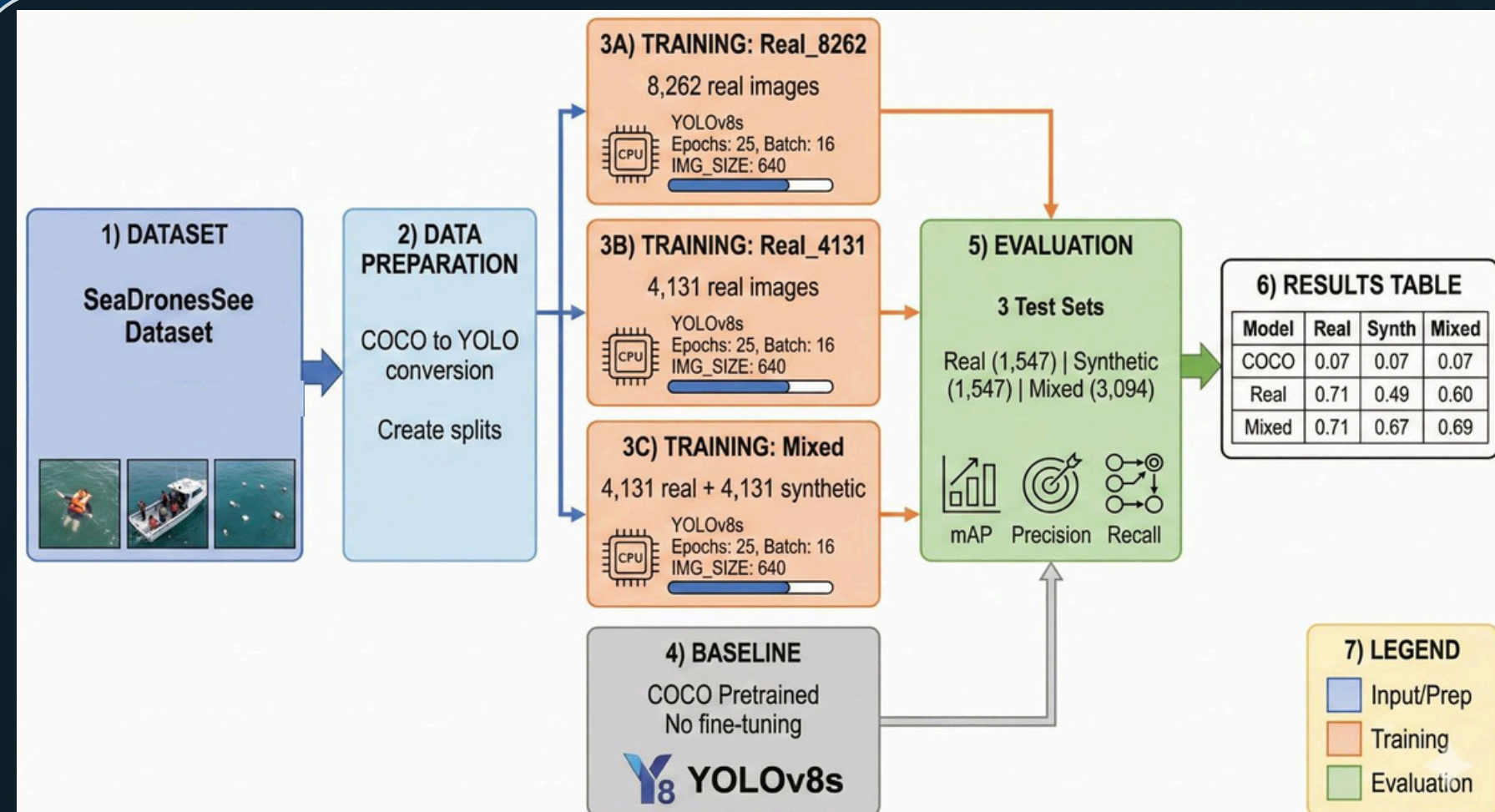
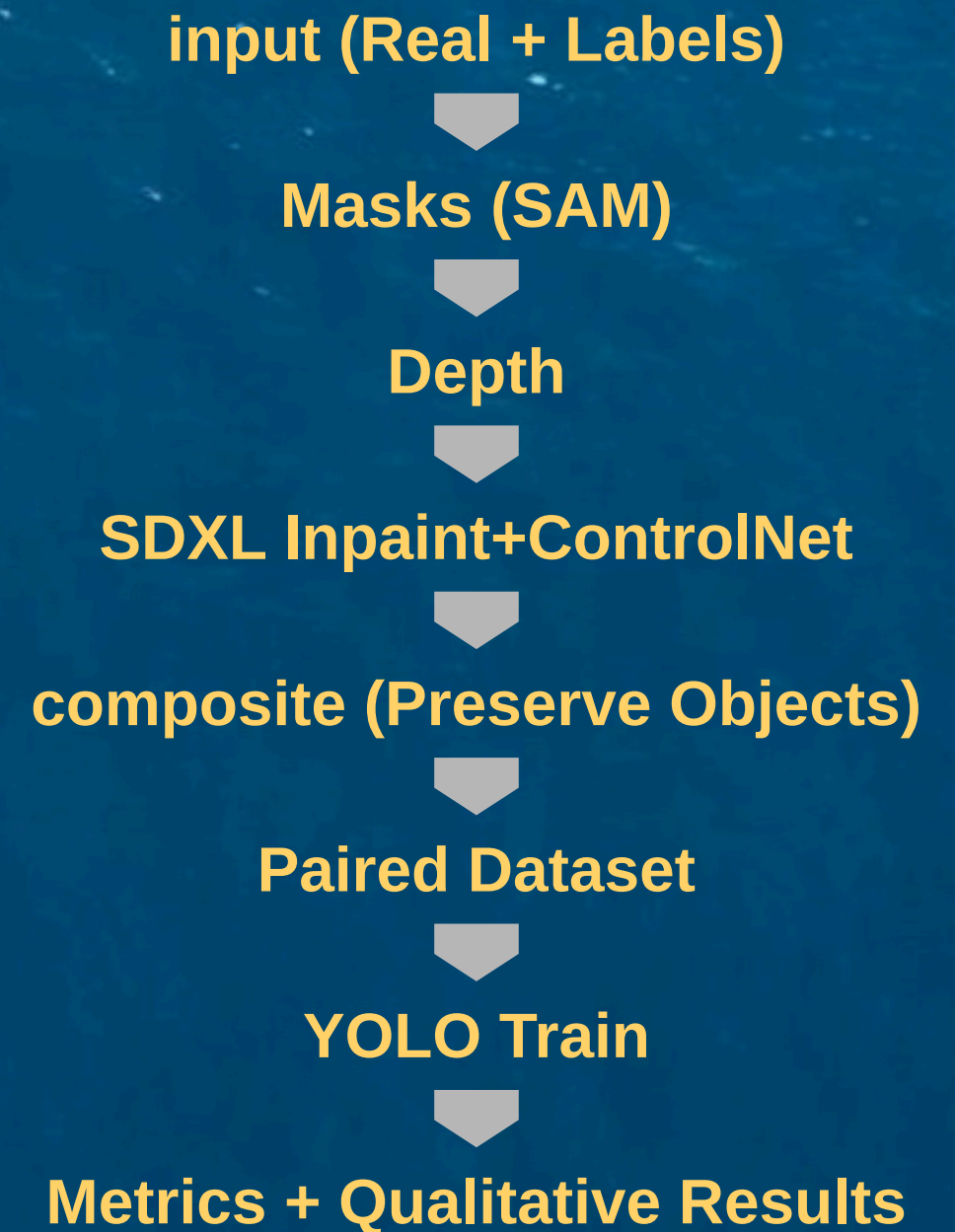
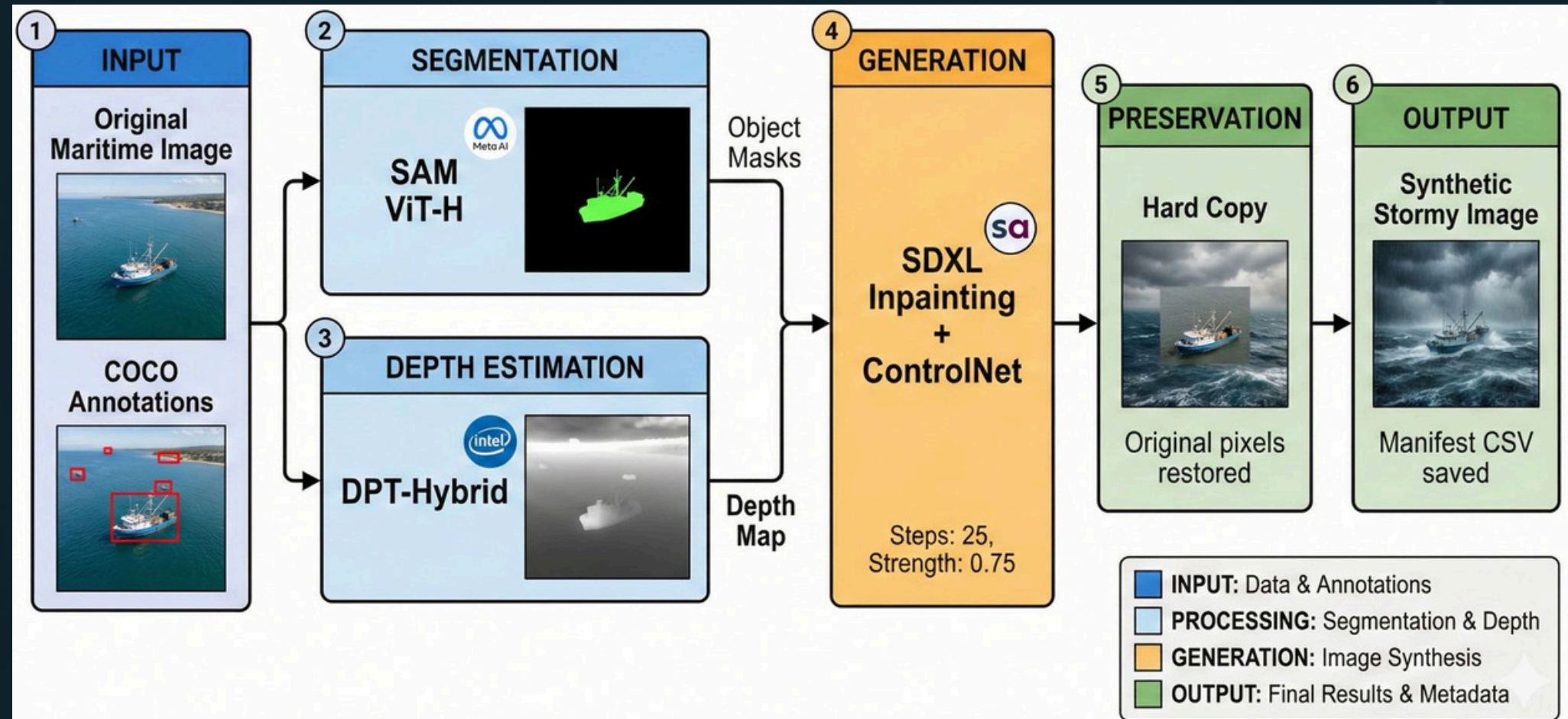
DELIVERABLES

- Robust YOLOv8 detector for stormy maritime conditions
- 5,678 synthetic images of stormy seas
- Comprehensive evaluation on 3 test sets (Real, Synthetic, Mixed)



The project's novelty lies in generating synthetic stormy sea backgrounds while perfectly preserving original objects, addressing the critical data gap in maritime SAR.

Review Methodology

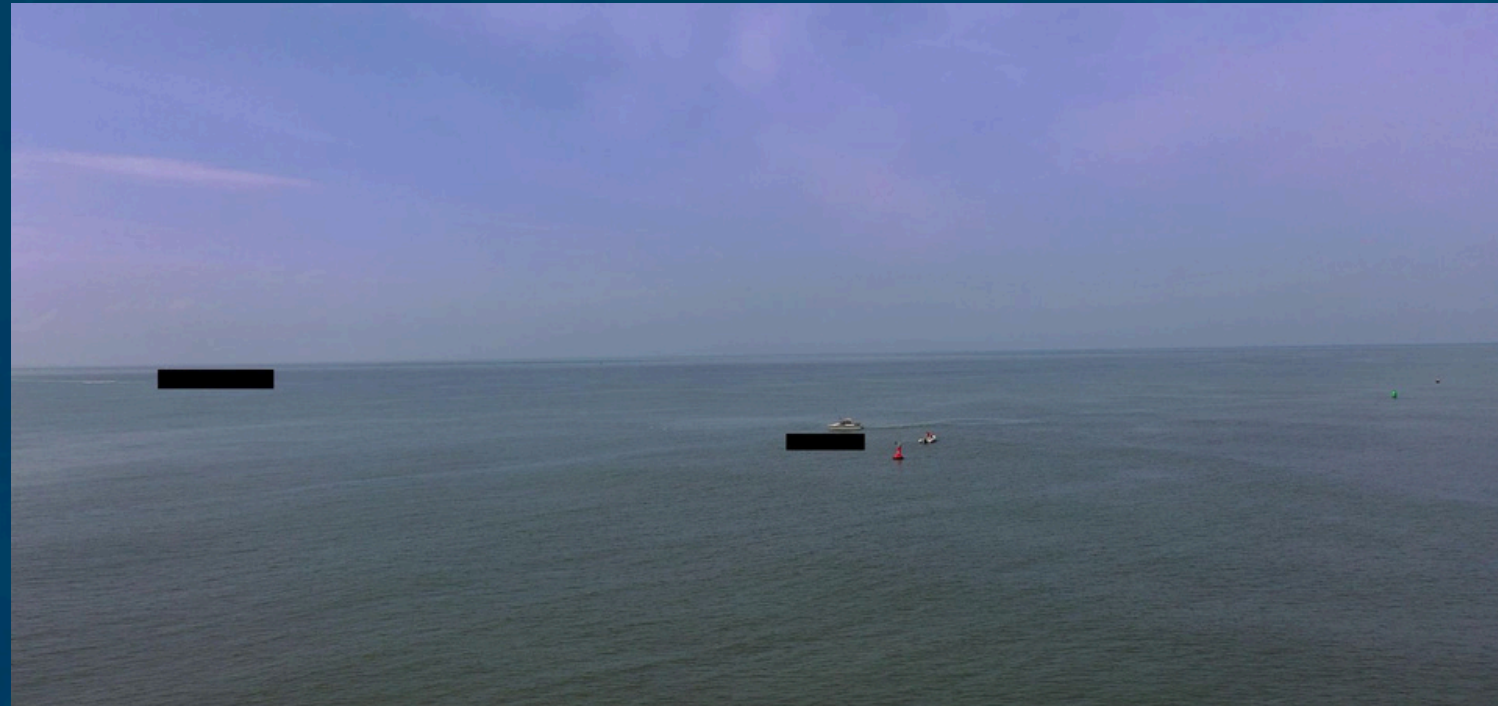


- Synthetic stormy background generation (SAM + SDXL + ControlNet)
- Object preservation via hard copy compositing
- Model training & evaluation (4 models)
- Robustness testing across 3 test sets

Examples

ORIGINAL IMAGE

SYNTHETIC IMAGE



Results

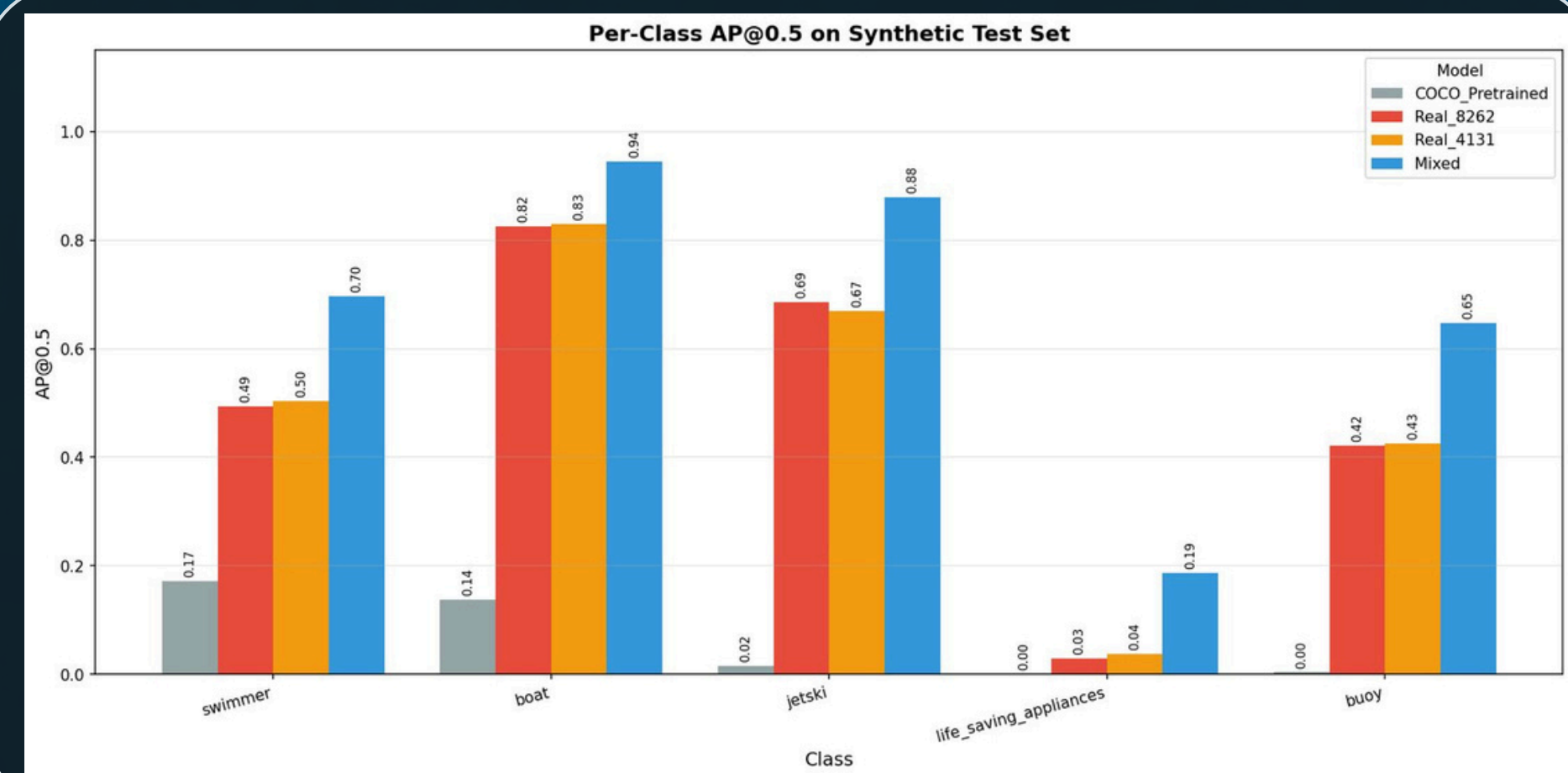
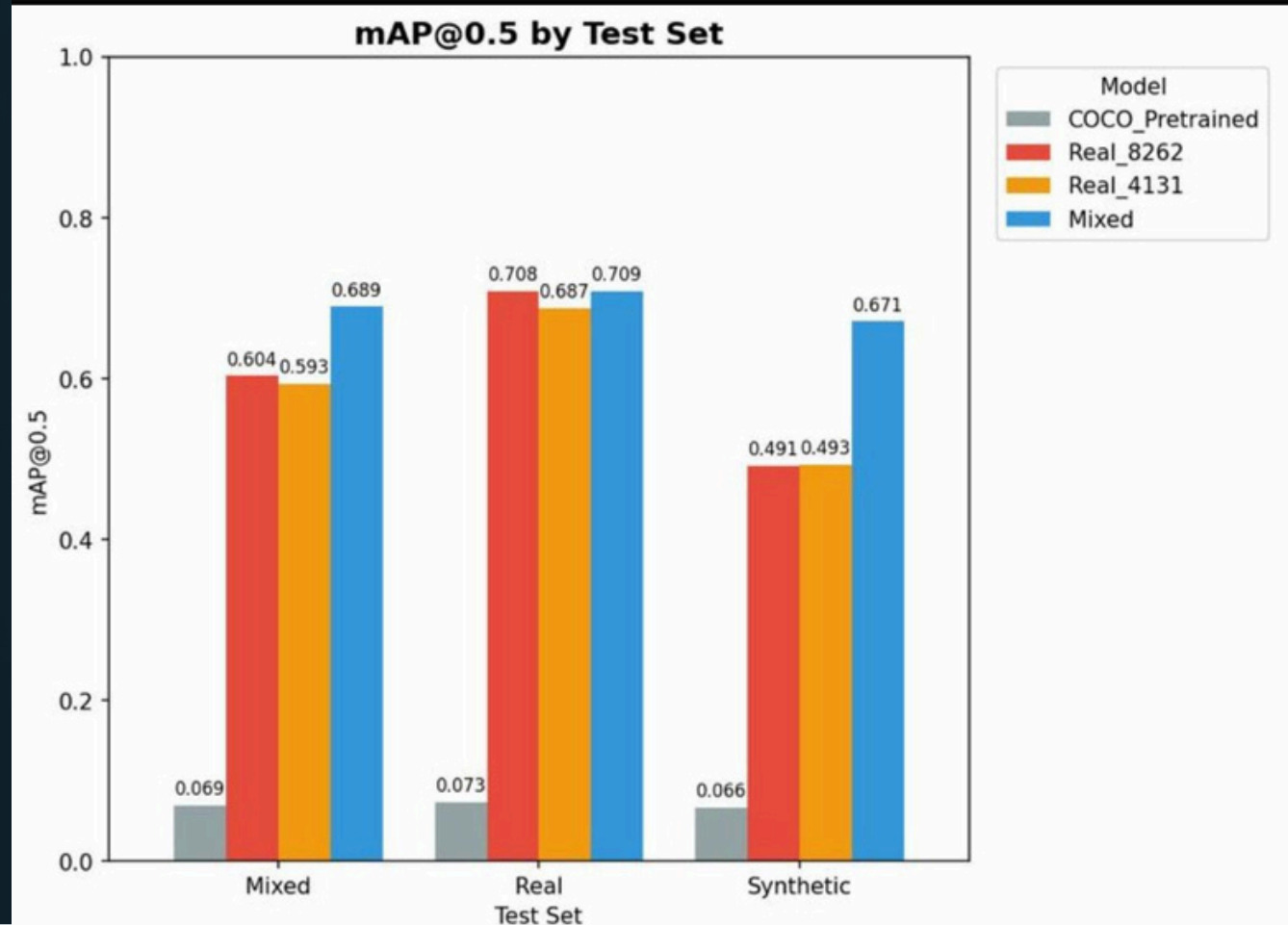
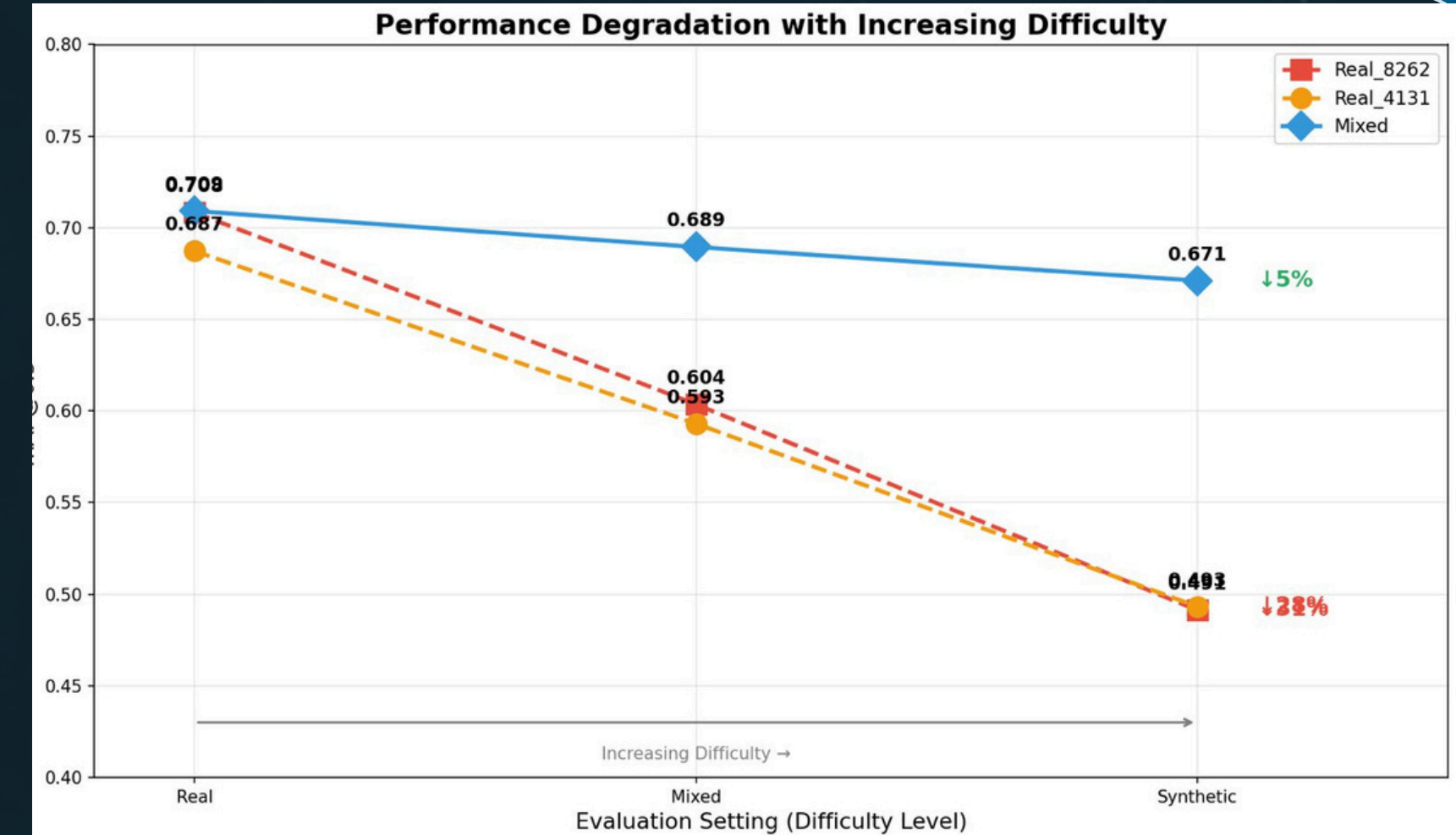
Model Selection:

Mixed (Real+Synthetic) achieves best robustness

Mixed Model Final Performance:

mAP@0.5 = 0.709 (real) mAP@0.5 = 0.671 (stormy)

- Synthetic augmentation dramatically improves storm detection



Conclusion & Future Work

✓ The pipeline successfully implements an end-to-end system for synthetic stormy sea data generation, enabling robust maritime object detection under adverse conditions.

✓ Improved detection robustness: Mixed model shows only 5% performance drop on stormy images vs 28% for Real-only models (6x improvement).

✓ Demonstrated effectiveness: Synthetic data augmentation achieves x10 improvement over COCO baseline and +18% on synthetic test vs fair comparison.

REMAINING CHALLENGES:

○ Generated storms may lack extreme wave physics accuracy

○ Very small objects (<20px) may have minor artifacts after compositing

IMPROVEMENT ACTIONS

- Improve wave realism with physics-based simulation
- Add more atmospheric effects (rain, fog, mist)
- Fine-tune SAM for maritime-specific segmentation
- Expand to night-time and low-visibility conditions
- Test on real stormy footage when available

ADDITIONAL EXPERIMENTS

- Try larger YOLO models (YOLOv8m, YOLOv8l)
- Experiment with different synthetic-to-real ratios
- Add temporal consistency for video processing
- Cross-dataset validation with other maritime datasets